

FinSight AI

Engineering Design Document (v1.0)

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Type: AI SaaS Platform

1. Product Vision

FinSight AI is an AI-powered Financial Intelligence Platform that unifies personal finance, investment research, portfolio analytics, and AI-powered reasoning into a single application. Users can upload financial documents, research companies, analyze portfolios, and receive citation-backed insights.

2. Goals

Primary: Financial document ingestion, AI Copilot, investment research, portfolio analytics, production-grade RAG.

Secondary: Demonstrate AI Engineering, Backend Engineering, Distributed Systems, Agentic AI, Observability, Evaluation.

3. Target Users

Retail Investors, Working Professionals, and Power Users requiring market intelligence and personal finance insights.

4. Functional Requirements

Authentication (JWT), document uploads (PDF/CSV/XLSX/DOCX), financial analytics, portfolio analysis, market intelligence, AI Copilot, report generation.

5. Non-Functional Requirements

99.9% availability, encrypted storage, audit logs, streaming responses, scalable architecture, observability, latency under 2 seconds for common queries.

6. High-Level Architecture

Frontend (Next.js) → API Gateway (FastAPI) → Auth, User, Document, Portfolio and AI Services → Celery/RabbitMQ → OCR & Parsing → Hybrid Retrieval (BM25 + pgvector + Cross Encoder) → LangGraph → LLM Gateway (OpenAI/Ollama/Claude/Gemini) → PostgreSQL, Redis, Object Storage, LangSmith, Phoenix.

7. AI Architecture

Planner Agent delegates work to Finance, Portfolio, Research and Document Agents. Outputs are merged by a Reasoning Agent, validated, cited, and streamed to the user.

8. RAG Pipeline

Upload → OCR → Cleaning → Metadata Extraction → Chunking → Embeddings → pgvector → BM25 → Hybrid Retrieval → Cross Encoder → Context Builder → LLM → Citation Verification.

9. AI Agents

Planner Agent, Finance Agent, Portfolio Agent, Market Research Agent, Report Generator Agent, Evaluation Agent.

10. Database Design

Core tables: Users, Documents, Chunks, Transactions, Portfolio Holdings, AI Conversations.

11. APIs

POST /upload, POST /query, POST /portfolio, GET /dashboard, GET /analytics, GET /research, POST /reports, POST /agent/chat.

12. AI Evaluation

Track retrieval precision/recall, context precision, faithfulness, answer relevance, latency, token usage, hallucination rate and cost.

13. Monitoring

Prometheus, Grafana, LangSmith and Phoenix for latency, tracing, prompt versions, queue depth, failures and cost tracking.

14. Technology Stack

Frontend: Next.js, React, Tailwind. Backend: FastAPI, PostgreSQL, Redis, Celery, Docker. AI: LangGraph, LangChain, pgvector, Sentence Transformers, Cross Encoder, OpenAI, Ollama. Observability: LangSmith, Phoenix, Prometheus, Grafana.

15. Roadmap

Phase 1: Infrastructure & Auth
Phase 2: Document Ingestion & RAG
Phase 3: Personal Finance Analytics
Phase 4: Market Intelligence
Phase 5: Portfolio Intelligence
Phase 6: Agentic AI
Phase 7: Observability & Evaluation
Phase 8: Production Deployment

16. Resume Highlights

Production-grade multi-agent AI platform, hybrid RAG, financial document processing, asynchronous backend, evaluation pipeline, local & cloud LLM deployment, observability, Dockerized architecture.